Leukemia Cell Classification Using Deep Learning Model

FYDP I Report

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Chapter 1

Introduction

1.1 Project Overview

Leukemia is a life-threatening blood cancer that affects the white blood cells of the bone marrow. It occur due to the uncontrolled production of white blood cells, which interfere with the immune system's ability to fight against infection and produce red blood cells. Leukemia needs early and accurate diagnosis to improve patient survival probability.

The traditional diagnosis of leukemia through manual microscopic examination is time consuming & also highly dependent on pathologists expertise, which may lead to delays and sometimes error. The aim of this project, Leukemia Cell Classification Using a Deep Learning Model, is to develop an automated system for leukemia classification using medical image analysis.

Blood smear slides are collected from various hospitals, and then, using a microscope with a high-resolution camera, microscopic images are captured from the slides. Medical experts label cells into five categories: healthy, Acute Lymphoblastic Leukemia (ALL), Acute Myeloid Leukemia (AML), Chronic Lymphocytic Leukemia (CLL), and Chronic Myeloid Leukemia (CML). Image preprocessing techniques are then applied to improve dataset quality and diversity. Then split the dataset for training, validation & testing. Data augmentation techniques are applied to improve dataset diversity & reducing overfitting.

ResNet, which is a deep learning model, is used for its ability to extract hierarchical and learn complex cellular features. After training and testing, this system enhances diagnostic accuracy and efficiency in the real healthcare environments through automated leukemia detection from microscopic images.

1.2 Motivation

Leukemia is a hematological malignancy that occurs due to the uncontrolled production of abnormal white blood cells in the bone marrow and peripheral blood. The World Health Organization (WHO) estimates that approximately 474,000 new cases and over 311,000 deaths occur annually due to leukemia. According to Blood Cancer United, approximately 66,890 people were expected to be diagnosed with leukemia in 2025 in the USA. They also stated that around 475,323 people are living with it. AACR Cancer Progress Report 2025 says that there are 66,890 new cases and 23,540 deaths in the US because of leukemia.

Early detection is very important because leukemia progresses very rapidly. This disease may develop different variants which reduce the effectiveness of treatment. Early detection allows for timely and targeted treatment that improves survival rates, reduces mortality rates and complications.

Manual or traditional detection is very time consuming and highly dependent on the expertise of hematologist's. In the early stage, symptoms and abnormalities are often to subtle to be detected through manual examination. Therefore, deep learning models can be effective in detecting leukemia in the early stage by learning past data.

1.3 Objectives

The main objective of our research is to develop a real-time, automated, and reliable system using deep learning architectures to classify leukemia subtypes (ALL, AML, CLL, CML) from microscopic blood smear images. The specific objectives of our research are:

- To develop an end-to-end automated leukemia classification framework based on deep learning techniques to minimize human error.
- To implement image preprocessing and synthetic data augmentation techniques to reduce class imbalance and improve model performance across diverse clinical datasets.
- To utilize Convolutional Neural Networks (CNN) for the automated extraction of high-dimensional and discriminative features from the blood smear images which are often too subtle to be detected by the human eye.
- To classify leukemia samples into specific subtypes (ALL, AML, CLL, CML) and separate them from normal blood samples with high accuracy.
- To validate the performance of ResNet using standard evaluation metrics including F1 score, recall, accuracy, precision, and confusion matrix to ensure clinical reliability.

- To analyze the feasibility and effectiveness of AI-assisted leukemia classification as a secondary tool for early detection of leukemia.
- To support hematologists in clinical decision making.

1.4 Methodology

This study aims to develop and evaluate a deep learning based automated system for leukemia detection. The proposed approach focuses on classifying microscopic blood smear images into leukemia and non-leukemia categories, as well as leukemia subtype classification (ALL, AML, CLL, CML).

Dataset Description

We have collected the dataset from various hospitals with following proper ethical guildelines and regulations. Peripheral Blood Smear (PBF) slides were collected and microscopic images were captured. Then the dataset was split into

- Training Dataset: 70%
- Validation Dataset: 15%
- Test Dataset: 15%

We tried to create the dataset in a way that we can dataset bias.

Data Preprocessing

In medical image processing this is one of the most crucial part because it helps enhance model performance.

1. Image Resizing: All images are resized to a fixed resolution to match the input requirements.
2. Noise Reduction: Techniques like Gaussian Blur are applied to minimize artifacts and enhance image clarity.
3. Normalization: Pixel intensity values are normalized to a standard range, ensuring consistent feature representation across the dataset.
4. Data Augmentation: applied to mitigate overfitting and class imbalance. Data augmentation techniques including rotation, flipping, zooming and brightness adjustment are used.

Model Architecture

The proposed system uses ResNet50 due to its effectiveness in medical image analysis. ResNet architecture consists of multiple convolutional layers for feature extraction.

1.5 Project Outcome

Our project aims to develop a system that can automatically classify leukemia cells from blood smear images by using a deep learning model. This model is expected to achieve strong performance in terms of accuracy, recall, precision, & F1 score.

This approach reduces the need for manual diagnosis, improves consistency and make the diagnostic process faster. The Proposed solution increases access for early leukemia detection. It also save time, improve diagnostic efficiency and provide clinical support in healthcare environment.

Chapter 2

Background

2.1 Preliminaries

This section establishes the foundational medical, computational, and mathematical concepts required for automated leukemia detection and classification.

2.1.1 Medical Foundations and Leukemia Taxonomy

Leukemia is a hematological disease caused by the abnormal production of White Blood Cells (WBC). These malignant cells originate in the bone marrow and interfere with the production of healthy blood cells.

Let

$$\mathcal{C} = \{c_1, c_2, \ldots, c_K\}$$

denote the set of classes in this study, where K = 5:

- c_1 (**Healthy**): Normal white blood cell morphology.
- c_2 (**ALL**) – Acute Lymphoblastic Leukemia. A fast growing cancer of immature lymphoid cells called lymphoblasts in the bone marrow. Common in children but can affect adults. Symptoms: fatigue, infections, bone pain, bleeding and fever.
- c_3 (**AML**) - Acute Myeloid Leukemia. It is also a fast growing cancer of immature myeloid cells. More common in adults and older people. Symptoms including anemia, infection and bleeding.
- c_4 (**CLL**) – Chronic Lymphocytic Leukemia. A slow growing cancer of mature B-lymphocytes which primarily affects older adults. Symptoms: fatigue, weight loss.
- c_5 (**CML**) – Chronic Myeloid Leukemia. It's also a slow growing cancer of myeloid

stem cells which associated with the Philadelphia chromosome. Symptoms including fatigue, weight loss, enlarged spleen and night sweats.

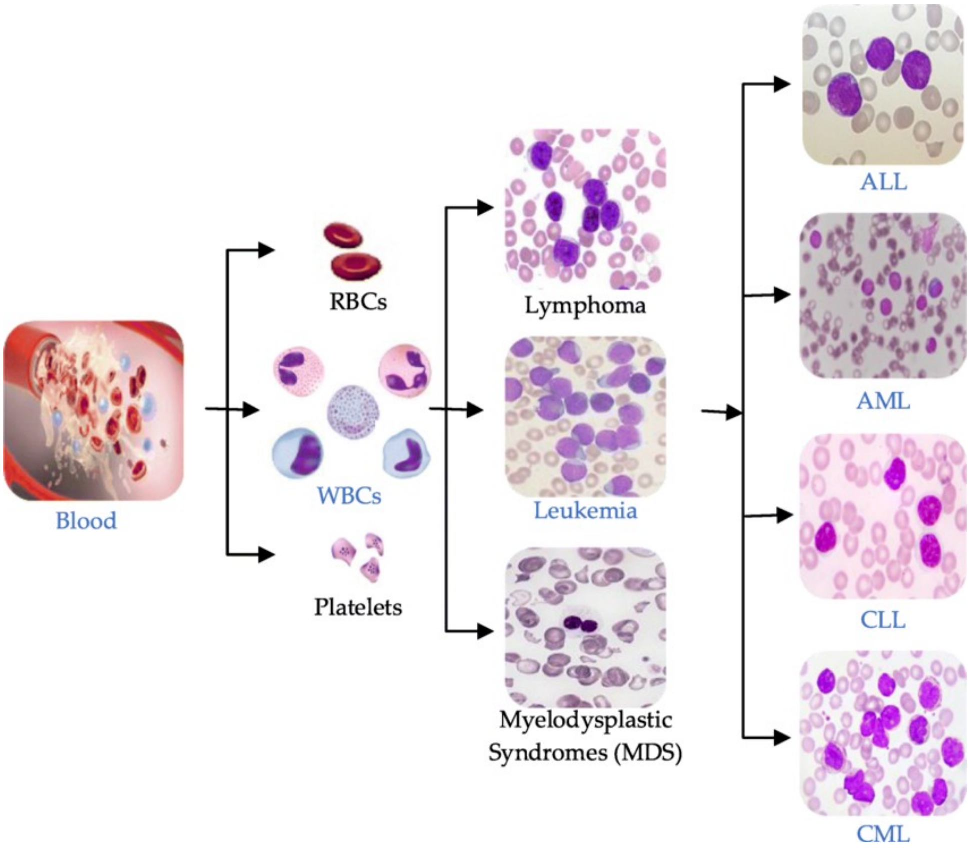

Figure 2.1: Subtype of Leukemia

Peripheral Blood Smear (PBS): PBS is the gold standard diagnostic tool in which blood is spread on a glass slide and stained. It enables visual analysis of WBC morphology, including cell size, nuclear shape, and cytoplasmic texture.

2.1.2 Image Preprocessing and Data Preparation

To enhance data quality and ensure model compatibility, a transformation $T(\cdot)$ is applied to raw images

$$X \in R^{H \times W \times 3}$$

such that:

$$X' = T(X)$$

Preprocessing: This includes resizing images to fixed input dimensions required by the model, Gaussian blurring for noise reduction and Contrast Limited Adaptive Histogram Equalization (CLAHE) to enhance nuclear details.

Data Augmentation: To mitigate overfitting in limited medical datasets, N synthetic samples are generated:

$$\tilde{X}_i = A_i(X), \quad i = 1, 2, \ldots, N$$

where $A_i(\cdot)$ represents stochastic transformations such as rotation, zooming and flipping.

Class Imbalance: Occurs when

$$N_{\text{healthy}} \gg N_{\text{leukemia}}$$

This imbalance is handled by using Synthetic Minority Over sampling Technique (SMOTE), random oversampling or class weighting w_k in the loss function.

2.1.3 Machine Learning Framework

The framework used Convolutional Neural Networks (CNNs) for automated feature extraction. The output of l-th convolutional layer can be expressed as:

$$\mathbf{F}^{(l)} = \sigma\left(\mathbf{W}^{(l)} * \mathbf{F}^{(l-1)} + \mathbf{b}^{(l)}\right)$$

where $*$ represents convolution, $\sigma(\cdot)$ is the activation function, and $\mathbf{F}^{(l)}$ represents the feature map.

Hierarchical Learning: CNNs gradually learn simple features like edge at early stages and more complex shape patterns which important for leukemia detection.

Transfer Learning: Model parameters θ are initialized using pre-trained weights (e.g., ResNet or MobileNet):

$$\theta = \theta_{\text{pretrained}} + \Delta\theta$$

This accelerates convergence and improves generalization.

Explainable AI (XAI): Gradient weighted Class Activation Mapping (Grad-CAM) is used to generate heatmaps highlighting regions of interest (ROI) that influence model predictions, enhancing clinical interpretability.

2.1.4 Loss Function and Performance Metrics

To address multi-class classification and class imbalance, the Weighted Categorical Cross-Entropy loss is used:

$$\mathcal{L} = -\sum_{k=1}^{K} w_k y_k \log(\hat{y}_k)$$

where y_k and $\hat{y}_k$ denote the ground truth and predicted probability for class k, respectively.

Model performance is evaluated using a confusion matrix comprising True Negatives (TN), True Positives (TP), False Negatives (FN) and False Positives (FP).

Table 2.1: Performance Evaluation Metrics

Metric	Formula	Description
Accuracy	$\frac{TP+TN}{TP+TN+FP+FN}$	Overall correctness
Precision	$\frac{TP}{TP+FP}$	Reliability of leukemia detection
Recall	$\frac{TP}{TP+FN}$	Sensitivity to leukemia cases
F1-score	$2 \cdot \frac{PR \cdot RE}{PR+RE}$	Balance between precision and recall
G-Mean	$\sqrt{\text{Sensitivity} \times \text{Specificity}}$	Performance on imbalanced data

2.1.5 Architecture Found

Support Vector Machine (SVM)

It is a supervised machine learning algorithm used for both Classification & Regression problems. It determines the best hyperplane that divides data points of different classes by maximizing the margin between them. The data sample that are closest to the hyperplane are known as support vectors, and they play an important role in decision boundaries.

SVM works well in leukemia classification, and it has been successfully used in many research papers.

Why SVM is suitable for Leukemia Classification

1. High Dimensional Data handling - Leukemia datasets have many features & SVM performs very well in high dimensional space
2. Perform well with small dataset - Medical datasets are maximum time limited in size & SVM can achieve high accuracy with small data
3. Kernel Flexibility - RBF kernel – used for leukemia image based classification – Linear kernel – good for gene expression data
4. Good Generalization - By maximizing the margin, SVM can reduce overfitting which important for medical diagnosis
5. Already prove in Research performance for
    - ALL vs AML
    - Normal vs Leukemia cells
    - Leukemia sub type classification

Steps of SVM for Leukemia classification

1. Input Data - Blood smear image or - Extracted feature like texture, shape, color of nucleus
2. Image preprocessing - Noise Removal - Data Augmentation - Feature Normalization - Segmentation of white blood cells
3. Feature Extraction - GLCM, HOG and LBP for feature extract from image - For deep feature extraction CNN+ SVM hybrid
4. SVM Classification - Used RBF kernel function
5. Evaluation - Accuracy, sensitivity - F1 score, Precision, Recall and confusion matrix

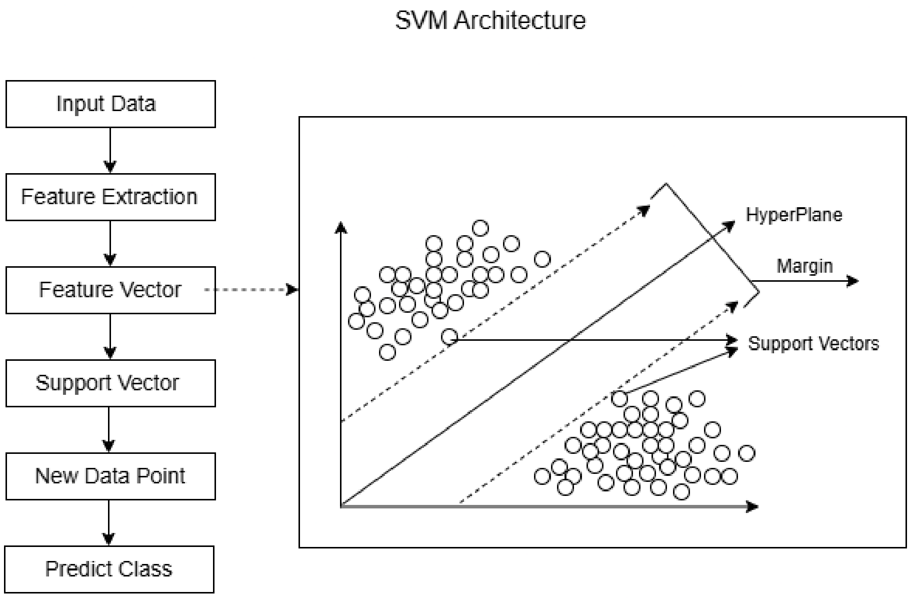

Figure 2.2: SVM Representation

Summary- SVM is good and effective for leukemia classification and detection, especially when the dataset is limited in size. With Proper feature extraction and RBF kernel function selection, SVM can achieve high classification accuracy in Leukemia detection and classification.

Convolutional Neural Networks (CNN)

CNNs are deep learning models specifically designed for image analysis. Since images are high-dimensional data, manual feature extraction is difficult. CNNs address this challenge by automatically learning important visual features (edges, shapes, textures, and cells) directly from raw images. Following picture represents a typical CNN architecture

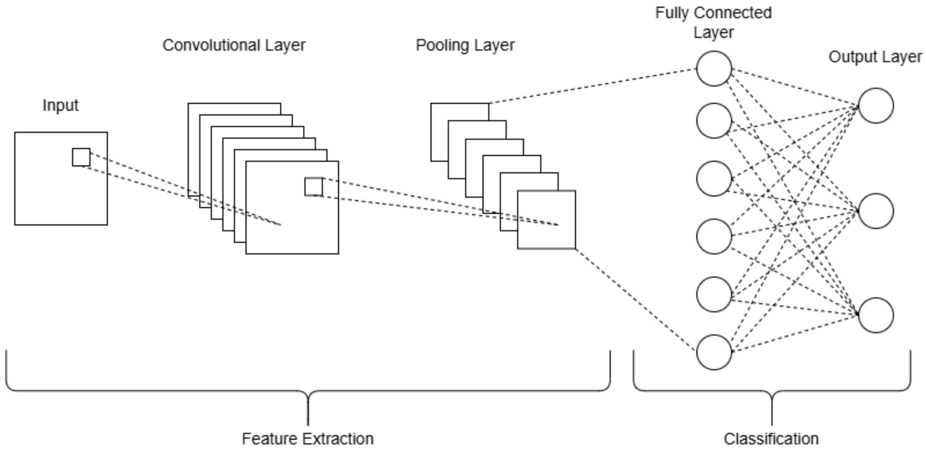

Figure 2.3: CNN Architecture

Convolutional Layers slide over images using small filters or kernels to capture specific feature or pattern such as edges, texture or more complex structure in deeper layers. Pooling Layers are used with convolutional layers to reduce the spatial dimension of the input which make it easier to process and requiring less memory while preserving the important information. Fully connected layers receive high dimensional outputs from the previous layers and combine and integrate all the extracted features across the entire image instead of focusing only on local regions. Finally, output layer generates the predicted class probabilities. However, CNNs are highly effective in medical image analysis because of their ability to learn complex visual patterns directly from raw images. This capability makes them suitable for automated leukemia detection and classification using microscopic blood smear images.

Decision Tree

It is a supervised learning algorithm commonly used for classification and regression tasks. It represents the decision making process using a tree like structure, where each internal node performs a test on a feature, each branch shows the result of that test and each leaf node give the final class label or predicted value. Because of its simple structure and easy interpretation, decision tree is commonly used as a baseline classifier in many research studies.

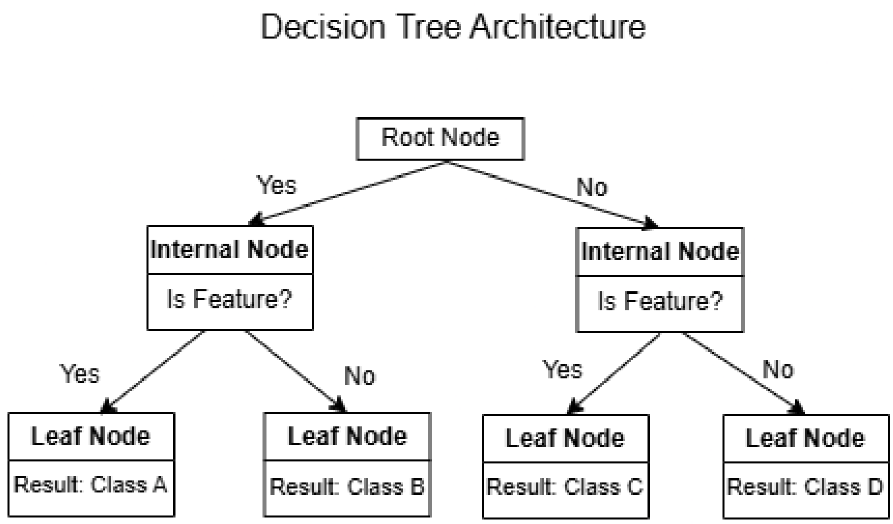

Figure 2.4: Decision Tree Architecture

Working Principle of Decision Tree: It works by repeatedly splitting the dataset into smaller groups based on feature values. At every node, it selects the feature that best separates the data into different classes. The choice is guided by Entropy and Information Gain. The splitting process continues until a stopping condition is satisfied such as reaching a maximum depth or having pure leaf nodes.

Decision Tree in Image Classification: In image classification tasks, raw images are usually converted into feature vectors using feature extraction methods such as texture, shape or color features. These extracted features are then used as input for the decision tree classifier. The model learns decision rules that help distinguish between different image classes based on these features.

Random Forest

It's a machine learning model based on an ensemble of multiple decision trees to make better predictions. It improves prediction accuracy and reduces error by combining many trees each with different criteria. Each tree is built using random subsets of the data. Their predictions are combined through voting for classification and averaging for regression which makes it an ensemble method.

Decision trees are helpful, but they can be prone to problems. These problems include:

1. Bias: This occurs when a degree of error is introduced into the model, often because the instance space isn't evenly split during training, meaning the model might not see all data points. Example-Imagine we are trying to learn about diff types of fruits but only ever look at apples. If we then try to identify a banana, we'd be very biased towards saying it's an apple because that's all we've seen. This "only

looking at apples" is like unevenly splitting instance space during training our model is only seeing a limited or skewed part of the overall data. Random Forest solves reducing bias by creating many decision trees. Every tree is trained on a different and random subset of main dataset. So, instead of just one learner seeing only apples- One tree might see apples and oranges. Another might see bananas and grapes. A third might see a mix of everything, but in a different order.

2. Variance: it happen when model become more sensitive to fluctuations in the training data, causing it to fit noise instead of the actual underlying pattern.

3. Overfitting: This happens when model start to memorize the data rather than trying to generalize for making predictions on future data. Example: Imagine studying for a test. Just memorize every single question and answer from a practice test Instead of understanding the concepts.

4. Training Data (Practice Test): When it takes the practice test again, it scores perfectly because it memorized all the answers.

5. New Data (Real Test): When it takes the actual test, the questions are slightly different, or phrased in a new way. Because it only memorized answers and didn't learn the underlying concepts, it performs poorly on the real test.

Random Forests common solutions to overfitting in machine learning -

1. Data-Based Solutions

   - Collecting More Data: The most effective method, as a larger dataset makes it harder for the model to memorize specific noise.
   - Data Augmentation: Expanding the original dataset by generating modified copies of existing data (e.g., rotating images, adding noise).
   - Feature Selection/Reduction:Eliminating unnecessary or repetitive features that may negatively affect the model.
   - Cleaning Data: Removing outliers or errors from the training set.

2. Model-Based Solutions

   - Reducing Model Complexity: Simplifying the model's architecture by using fewer layers or neurons in neural networks.
   - Regularization: Adding penalty terms into the loss function to limit excessive model complexity (e.g., L1/L2 regularization).
   - Dropout: In neural networks, randomly turning off a portion of neurons during training to reduce dependency on specific connections.

3. Training-Based Solutions

    - Cross-Validation: Test the model performance on multiple subsets of the dataset to check its robustness and generalization ability.
    - Early Stopping: Observing the performance of the model on a separate validation set and stop the training process when performance stops improving even if training error is still decreasing.
    - Hyperparameter Tuning: Adjust model settings to find the best balance between complexity and accuracy.

Use of Random Forest

1. Complex Data: Large datasets with many features.
2. High Accuracy Robustness: When a model is required that is less likely to overfit than a single decision tree.
3. Classification Regression: Can predict categories like spam detection, disease diagnosis and numerical values like stock prices, customer satisfaction.
4. Non-linear Patterns: Good at capturing complex relationships that are not straight line trends.
5. Feature Importance: Useful for identifying key features in data, though caution is needed when features are correlated.
6. Handling Missing Data: Can effectively estimate missing values.

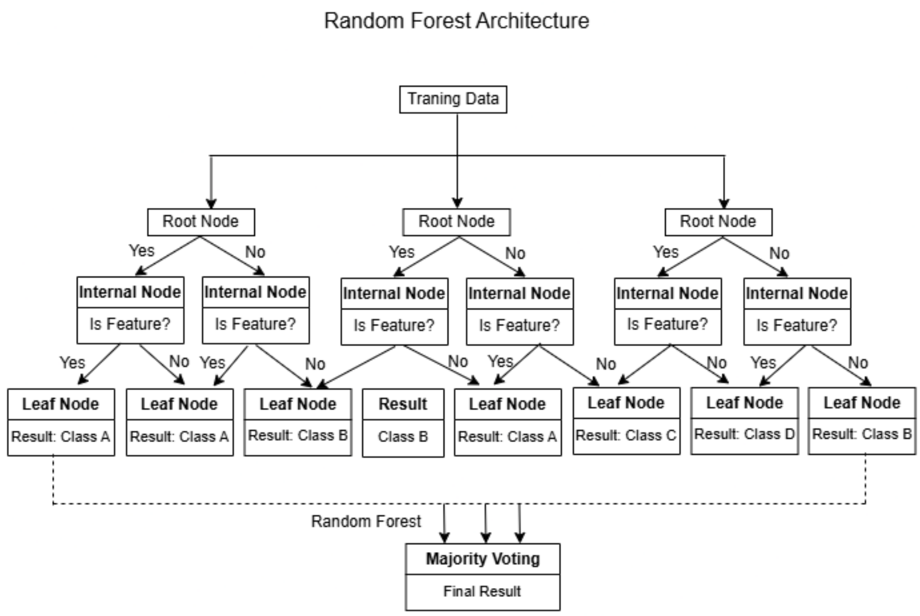

Figure 2.5: Random Forest Architecture

Random Forests reduce these problems by averaging multiple decision trees, which lowers variance and improves robustness.

K-Nearest Neighbors(KNN)

KNN algorithm is a straight forward, non parametric and supervised learning model used for classification and regression tasks. It predicts the label of a new data point based on the labels of its nearest neighbors.

How the KNN Algorithm Works The core idea of KNN is that similar data points are located near one another. To classify or predict a new data point, the model follows these steps:

1. Choose the number of neighbors (K): Select a positive integer value for K. It suggested to choose an odd number for K for binary classification to prevent ties.

2. Compute distance: Measure the distance between the new data point and all existing data points in the dataset using metrics like Euclidean distance and Manhattan distance. 3. Identify the K nearest neighbors: Selects the K data points that are closest to the new point. 4. Assign a class (or value):

   (a) For classification: The new data point is assigned to the class that appears most frequently among its K neighbors (majority vote).

   (b) For regression: Predicted value is calculated as the average (or weighted average) of the values of the K nearest neighbors.

Key Characteristics

- Supervised Learning: KNN need a labeled dataset to train on.

- Non-Parametric: It does not assume any specific distribution for the data.

- Lazy Learner (Instance-Based): KNN does not build a model during training. It stores the dataset and performs calculation only when making prediction.

Applications

KNN is a versatile algorithm used in various fields:

- Recommendation Systems: Companies like Netflix and Amazon use it to suggest products or movies based on similar users' preferences.

- Healthcare: It can be used in predicting health risks like heart attacks or cancer based on gene expressions.

- Finance: It helps with stock market forecasting and fraud detection.

- Image Recognition: It is used in handwriting and image recognition tasks.

Limitations

Despite its simplicity, KNN has drawbacks:

- Computational Cost: As a lazy learner, storing all data points makes it memory-intensive and computationally expensive for large datasets.
- Curse of Dimensionality: The performance of KNN decreases in high-dimensional spaces where distance calculations become less reliable.
- Feature Scaling: KNN is sensitive to the scale feature, so its important to scale all feature to ensure each one contributes equally to the distance calculation.

2.2 Literature Review

2.2.1 Related Work

Automated leukemia detection and classification from microscopic blood smear images has been widely studied using traditional machine learning and more recently deep learning techniques. Early research mainly focused on manual feature extraction combined with classical classifiers, while recent studies strengthen deep CNN and transfer learning to improve accuracy and robustness.

Multiple review based studies establish the medical necessity for automated leukemia diagnosis. Comprehensive reviews on ALL highlight that traditional diagnostic methods like manual microscopy are labor intensive and mostly can happen human errors. These studies didn't gave any model or report accuracy metrics but identified diagnostic inconsistencies as major bottlenecks.

Computational approaches relied on handcrafted features such as cell size, shape, color, texture etc. Studies using multi layer perceptrons trained with algorithms such as Bayesian Regularizations achieved almost 95% accuracy on binary classifications. But they need professional equipment and large investment for that research. Other works applied classifiers like SVM, K-NN, Naive Bayes and DT on extract feature. They achieve quite a good result but they struggled with class imbalance, feature sensitivity and multi class leukemia classification.

Deep learning for leukemia classification

Most common deep learning model used in leukemia detection is CNN. By using datasets like ALL-IDB and ASH Image Bank, CNN-based models achieved almost 88% accuracy for binary classification and 81–82% accuracy for multi-class classification. These models use image preprocessing steps such as resizing, normalization, and contrast enhancement and automatic feature extraction through convolutional layers.

Handling Imbalance data overfitting and underfitting

Imbalance dataset is a common problem in leukemia detection because most of the time abnormal cells are not identified or labeled. In some papers they used so many techniques to overcome this problem like SMOTE, under sampling and extensive data augmentation. Noise-filtered under sampling approaches improved minority-class recognition by enhancing F-measure and G-Mean, though they did not directly focus on deep learning models. More recent deep learning approaches tackle imbalance at the loss-function level. weighted loss functions, regularization, and dropout are some techniques used in different papers. These techniques reduce model bias toward majority classes.

2.2.2 Key Findings & Gap Analysis

Table: Key Findings & Gap Analysis .

Area	**Key Findings**	**Tools and Techniques Used**	**Gap**	**Proposed Solution**
Medical image-based detection of Acute Lymphoblastic Leukemia	MobileNetV2 achieved 99.69%, CNN 98.6%. SMOTE handled imbalance.	Dataset: ALL image dataset. Models: MobileNetV2, Custom CNN. Data augmentation.	Limited dataset, focused only on ALL, static dataset testing.	Collect larger multi-hospital dataset. Classify all leukemia subtypes.
Image analysis for detecting leukemia subtypes	CNN: 88% (ALL), 81% (multiclass). CNN better than traditional ML.	Datasets: ALL-IDB, ASH Image Bank. OpenCV, Keras. SGD, Adam. 5-fold CV.	Small dataset, multiclass lower accuracy, augmentation noise.	Collect larger dataset. Advanced augmentation. Apply deep learning models.
Biomedical image analysis for leukemia detection	MLP(BR): 94.51%, MLP(LM): 94.39%. 32 features best result.	Dataset: 800 images. Feed-forward MLP. Preprocessing. BR, LM algorithms.	Small dataset, single hidden layer, manual microscopy error.	Use deeper MLP, data augmentation, improve feature fusion.
Microscope image analysis using deep learning	Optimized CNN: 96% accuracy. Better than ResNet50 and YOLOv8.	3256 PBS images. Deep CNN (13 conv layers). Data augmentation.	Overfitting, black-box problem, limited to ALL & AML.	Collect more hospital data. Use Adam optimizer. Systematic augmentation.
Biomedical image classification using ResNet	ResNet better than CNN on large datasets. Transfer learning effective.	Datasets: ImageNet, COCO, Kinetics-400. ResNet, SGD, augmentation.	Ethical issues, high memory demand, limited attention mechanisms.	Add attention modules. Memory-efficient architecture. Bias detection workflow.
Computer-aided classification into FAB subtype	PCA NN achieved 86.4%. Good multiclass performance.	180 images. PCA-based NN. Preprocessing, feature extraction.	Small dataset, no augmentation, no DL methods.	Expand dataset. Use augmentation, cross-validation, deep learning models.

Area	Key Findings	Tools and Techniques Used	Gap	Proposed Solution
Cell classification using Deep CNN	ConvNet: 98.3%, AUC 0.99. Robust to translation error.	Herlev, HEMLBC datasets. Pre-trained ConvNet. 5-fold CV.	Slow inference, limited labeled data.	Use Fully CNN for automatic detection. Train with diverse cells.
Systematic description of leukemia & CAD	Highlights need for automation in diagnosis.	Medical imaging, CAD framework. Literature review.	No validation metrics. Focused more on medical side.	Develop AI-based CAD system. Promote computational research.
WBC classification using CNN	90% accuracy. Robust to noise.	CNN with preprocessing. Automatic feature learning.	Focused only on WBC, not leukemia subtype.	Extend model to classify leukemia subtypes.
Data enhancement with ResNet	Accuracy: 95%, F1: 93%. Effective multiclass classification.	Expert labeling. Preprocessing (CLAHE, resizing). Transfer learning.	Limited dataset, no multi-center validation.	Collect multi-center dataset. Advanced augmentation.
Noise filtration under-sampling	Improved F-measure and G-Mean.	Nearest neighbor filtering, distance-based under-sampling.	No deep learning. No subtype classification.	Apply deep learning model for leukemia classification.
Comparative evaluation of DL models	90% accuracy with imbalance handling.	DCNN, weighted loss, dropout, regularization.	No novel model proposed.	Optimize model selection for leukemia classification.
Loss function enhancement	98.92% accuracy with faster convergence.	Deep learning with optimized loss function.	Indirect leukemia detection. Focused on WBC.	Extend optimized loss for leukemia subtype classification.
ALL Segmentation using Dual Threshold	Improved ALL cell identification.	SVM, ANN, K-means, FCM.	No deep learning, small dataset.	Use deep learning with larger dataset.
CAD system for leukemia detection	89.8% accuracy with combined features.	GLCM, SVM, texture + shape features.	Limited accuracy, no deep learning.	Combine features with deep learning model.

Area	Key Findings	Tools and Techniques Used	Gap	Proposed Solution
Computer-Based Leukemia Diagnosis	KNN best accuracy 86%.	260 images. KNN, DT, SVM, Naive Bayes.	Small, low diversity dataset.	Collect larger dataset. Use advanced deep learning models.
Leukemia subtype detection using CNN	88.25% (binary), 82% (multiclass).	ALL-IDB, ASH Image Bank. OpenCV, Keras. SGD, Adam.	Binary better than multiclass. Model optimization needed.	Hybrid deep learning for better multiclass accuracy.
Classification of leukemia cell by CNN	96.6% accuracy for binary classification.	ALL-IDB1 dataset. MATLAB implementation.	Focused only on ALL. Limited real-world validation.	Optimize CNN. Validate on clinical data.
SVM for cancer tissue classification	SVM effective for cancer classification.	SVM, jackknife CV, dimensionality reduction.	Struggles with high-feature samples.	Develop systematic SVM training method.
Unintentional leukemia detection with CBC	Incidental detection risk of overdiagnosis.	CBC, clinical decision aids.	No systematic screening recommendations.	Implement active surveillance protocols.
Computer Vision & GANs	GAN enables image translation.	CycleGAN, PatchGAN, consistency loss.	Poor geometric transformation handling.	Hybrid supervised + GAN framework.
Spatial & technical data processing	Deep networks improve performance.	Multi-scale training, 3x3 filters, 1x1 conv.	High computational cost.	Optimize network depth. Reduce computation cost.
Deep learning for leukemia detection	Hybrid model achieved 97.04% accuracy.	Hybrid CNN + CCA + Bagging. SVM, KNN.	Small dataset.	Use parallel CNN hybrid approach.
Hybrid approach for ALL classification	ResNet50, SVM, and RF have the best performance.	ResNet152, VGG16, DenseNet121, MobileNetV2, and EfficientNet.	Insufficient data for deep learning. Manual detection is slow.	Use a hybrid deep feature selection approach.

2.3 Summary

From all these previous works, we identified some gaps in leukemia classification where mostly used CNNs and transfer learning, here main problem is small data set and single-centered and mostly focus on limited subtype like AML, ALL. Here they faced some common problems like class imbalance, overfitting and lack of pathological validation. Though Resnet based models work better than CNNs, some of them used them without attention mechanisms and robust imbalance handling. In our research we will try to mitigate these problems where we will enhance the Resnet model, data augmentation, handle class imbalance and overfitting and mainly we will label it by the Expert pathologist. This approach aims to improve multi class leukemia classification with more reliability and accuracy.

Chapter 3

Project Design

3.1 Requirement Analysis

The Main objective of our project is to classify leukemia cells from microscopic blood smear images using deep learning techniques. The system is designed to classify healthy, Acute Lymphoblastic Leukemia (ALL), Acute Myeloid Leukemia (AML), Chronic Lymphocytic Leukemia (CLL), Chronic Myeloid Leukemia (CML) cells to support early disease diagnosis.

3.1.1 Functional Requirements

1. We need blood smear slides from various hospitals & capture high quality microscope images from blood smear cells as input.

2. A medical expert is required to labeled image into five categories: healthy, ALL & AML, CLL, CML cells

3. Then apply image preprocessing steps like resizing, noise detection and remove, normalization & contrast enhancement.

4. Apply data augmentation techniques to increase dataset size because the dataset is limited.

5. Then apply machine and deep learning model like ResNet, SVM, Decision Tree, KNN and Random Forest for feature extraction and classification of leukemia. The model will produce evaluation metrics such as accuracy, precision, recall and F1 score.

3.1.2 Non-Functional Requirements

We need to achieve high classification accuracy above 90% on unseen data. To process the medical dataset efficiently, the system requires standard GPU-enabled system for the model training and testing.

3.2 Methodology

3.2.1 Project workflow

This section represent the overall workflow of our project, starting from data collection to final prediction. Each step of the workflow is explained below:

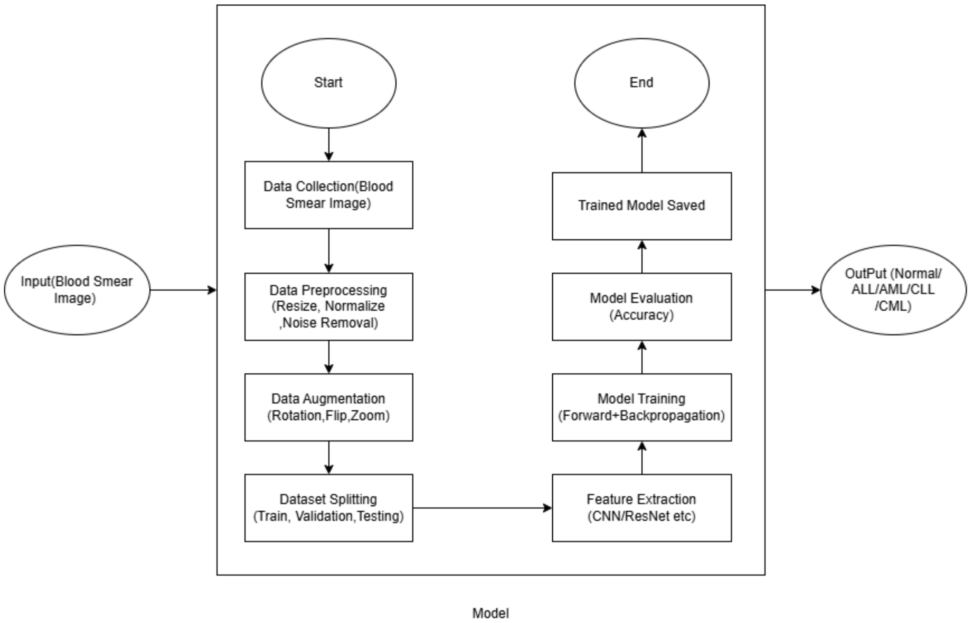

Figure 3.1: Working Flowchart

1. **Input – Blood Smear Image:**

   Project starts with the microscopic images of peripheral blood smear (PBS) slides which are captured using a high resolution digital microscope. Each picture contains White blood cells (WBCs) which may either have healthy or leukemia affected cells (ALL, AML, CLL, CML). This image is the primary input of the automated leukemia classification system.

2. **Data collection**

   Blood smear slides are collected from hospitals and diagnostic centers with proper ethical concerns. Images are captured with a high resolution digital microscope. Then all the images are labeled by expert hematologists into the following categories:

(a) Healthy

(b) Acute Lymphoblastic Leukemia (ALL)

(c) Acute Myeloid Leukemia (AML)

(d) Chronic Lymphocytic Leukemia (CLL)

(e) Chronic Myeloid Leukemia (CML)

3. **Data Preprocessing**

   Medical image preprocessing is a challenging step to enhance model performance, ensure consistency, improve image quality and reduce variability. The following preprocessing techniques are applied:

   (a) Image Resizing: All images are resized to a fixed resolution to match the input size required for the deep learning model

   (b) Noise Removal: Gaussian blurring or flirting techniques used for removing unnecessary background noise.

   (c) Normalization: Image pixel intensity values are scaled to a standard range from 0 to 1 to ensure stable and faster convergence during training.

4. **Data Augmentation**

   Medical datasets are always limited. Although we try to collect maximum data, but it is not enough, so we applied data augmentation to increase the dataset. The following data augmentation techniques are applied:

   (a) Rotation

   (b) Zooming

   (c) Brightness adjustment

   (d) Flipping

   Data augmentation helps to reduce overfitting, improve generalization and balance minority classes.

5. **Dataset splitting**

   The dataset is divided into three subsets:

   (a) Training Dataset: 70% used to train the model

   (b) Validation Dataset: 15% used to validate the model and prevent overfitting

   (c) Test Dataset: 15% used to evaluate model performance on unseen data through different metrics

Proper splitting ensures an unbiased evaluation and better generalization

6. **Feature Extraction**

   Now, a deep learning model ResNet are applied to automatically extract high level features from blood smear images. ResNet learns:

   (a) Edges and textures

   (b) Cell morphology patterns

   (c) Subtle visual differences between leukemia subtypes

   (d) Nuclear structure and cellular characteristics

7. **Model Training (Forward and Backpropagation)**

   During training:

   (a) Input images pass through the network during forward propagation

   (b) The model generates predictions for each leukemia subtype

   (c) Loss function calculates the error rate using weight and cross entropy

   (d) Backpropagation updates model weight and biases rate using ADAM or SGD

   This step continues for multiple iterations until the model performance satisfied.

8. **Model Evaluation**

   After training, the model is evaluated using the test dataset. Performance is evaluated using the following metrics:

   (a) Accuracy

   (b) Recall

   (c) Confusion matrix

   (d) Precision

   (e) F1 score

   These metrics ensure whether the model meets clinical standards

9. **Trained Model Saved**

   When satisfactory model performance is achieved, the trained model is saved for deployment. The saved model can be used later for real time leukemia classification without retraining.

10. **Output**

    Finally, when a new blood smear image is provided, the trained system predicts the image as following classes:

    (a) Healthy
    (b) ALL
    (c) AML
    (d) CLL
    (e) CML

    The output can serve as a clinical decision support tool to help hematologists in early leukemia detection

3.2.2 Pseudo Code of Algorithm

Algorithm 1 Leukemia Detection: Training Phase

Require: Blood smear image dataset D
Ensure: Trained model M

Step 1: Data Preprocessing
for each image $I \in D$ **do**
    $I \leftarrow$ Apply_CLAHE(I)
    $I \leftarrow$ Stain_Normalization(I)
    $I \leftarrow$ Noise_Removal(I)
    $I \leftarrow$ Cell_Segmentation_U_Net(I)
end for
Step 2: Data Augmentation & Class Balancing
$D_{aug} \leftarrow \emptyset$
for each image $I \in D$ **do**
    $D_{aug} \leftarrow D_{aug} \cup \{$Rotate($I$), Flip($I$), Zoom($I$)$\}$
    $D_{aug} \leftarrow D_{aug} \cup \{$MixUp($I$), CutMix($I$)$\}$
end for
$D_{bal} \leftarrow$ Apply_SMOTE(D_{aug})
$D_{bal} \leftarrow$ Generate_Synthetic_Samples_GAN(D_{bal})
Step 3: Dataset Splitting
$(D_{train}, D_{val}, D_{test}) \leftarrow$ Split(D_{bal})
Step 4: Model Initialization
$M \leftarrow$ ResNet50 (Pretrained)
Replace final layer of M with 5-class classifier
Step 5: Loss Function
Define $L(y, \hat{y}) =$
    $\alpha\cdot$ Focal_Loss$(y, \hat{y})$ $+\beta\cdot$ MCC_Loss$(y, \hat{y})$ $+\gamma\cdot$ Dice_Loss$(y, \hat{y})$
Step 6: Hyperparameter Optimization
$(\alpha, \beta, \gamma, \theta) \leftarrow$ Optimize_Using_Optuna()
Step 7: Model Training
Initialize M with θ
for epoch $= 1$ to N **do**
    for each batch $B \in D_{train}$ **do**
        $\hat{y} \leftarrow$ Forward(M, B)
        loss $\leftarrow L(B_{labels}, \hat{y})$
        Backpropagate(loss)
        Update_Weights(M)
    end for
    Validate(M, D_{val})
end for
return M

Algorithm 2 Leukemia Detection: Evaluation and Prediction Phase

Require: Trained model M, test dataset D_{test}
Ensure: Predicted class label $\in$ {Normal, ALL, AML, CLL, CML}
Step 1: Model Evaluation
Evaluate M on D_{test} using Accuracy, F1-score, MCC
Step 2: Explainability
Generate_GradCAM(M, D_{test})
Step 3: Prediction
for each new image I_{new} **do**
    $y_{pred} \leftarrow$ Predict(M, I_{new})
end for
Step 4: Output Mapping
for each y_{pred} **do**
    if $y_{pred} = 0$ **then**
        label $\leftarrow$ Normal
    else if $y_{pred} = 1$ **then**
        label $\leftarrow$ ALL
    else if $y_{pred} = 2$ **then**
        label $\leftarrow$ AML
    else if $y_{pred} = 3$ **then**
        label $\leftarrow$ CLL
    else if $y_{pred} = 4$ **then**
        label $\leftarrow$ CML
    end if
end for
return label

3.2.3 ResNet using Feynman's Path Integral

This section presents a physics-inspired interpretation of ResNet using the Feynman path integral formulation. Instead of viewing ResNet as a simple stack of nonlinear layers, we interpret it as a superposition of exponentially many computational paths, each associated with an energy functional.

Residual Learning as a Discrete Dynamical System

A standard residual block updates the feature representation as follows:

$$u(x, t + 1) = u(x, t) + F(u(x, t); W_t) \tag{3.1}$$

where $u(x, t)$ denotes the feature map at depth t, $F(\cdot)$ represents the residual mapping, and W_t are the learnable parameters of the t-th block.

In practical architectures, the residual mapping is defined as

$$F(u_t) = W_{2,t}\sigma(W_{1,t}u_t) \tag{3.2}$$

where σ denotes a nonlinear activation function (e.g., ReLU).

This formulation can be interpreted as a forward Euler discretization of:

$$\frac{\partial u}{\partial t} = F(u, t) \tag{3.3}$$

Thus, network depth corresponds to discrete time evolution of a dynamical system.

Network Architecture

The architecture follows a standard deep residual design with skip connections.

Input Stage

- Initial convolution (7×7, stride 2)
- Batch Normalization
- ReLU activation
- Max pooling

Residual Stages A typical configuration is shown in Table 3.1.

Table 3.1: ResNet Architecture Configuration

Stage	Number of Blocks	Output Channels
Stage 1	3	64
Stage 2	4	128
Stage 3	6	256
Stage 4	3	512

Each residual block consists of:

1. Convolution (3×3)
2. Batch Normalization
3. ReLU
4. Convolution (3×3)
5. Batch Normalization
6. Identity skip connection
7. Element-wise addition
8. ReLU activation

Path Integral Interpretation

Due to skip connections, information can propagate through multiple combinations of identity and residual mappings.

For a network with N residual blocks, each block provides two possible transitions:

- Identity (skip path)
- Residual transformation

Thus, the total number of possible computational paths is:

$$2^N \tag{3.4}$$

Each path p can be represented as:

$$p = (s_1, s_2, ..., s_N) \tag{3.5}$$

where

$$s_t = \begin{cases} 1, & \text{identity path} \\ F_t, & \text{residual path} \end{cases} \tag{3.6}$$

The contribution of a specific path is

$$u_{\text{path}} = u_0 \prod_{t=1}^{N} s_t \tag{3.7}$$

Therefore, the final output becomes:

$$u(x_N) = \sum_{p \in \mathcal{P}} u_0 \prod_{t=1}^{N} s_t \tag{3.8}$$

Hamiltonian and Energy Functional

To incorporate the physics-based interpretation, an energy (Hamiltonian) is assigned to each path:

$$H(p) = \frac{1}{2}(\sigma^2 p^2 - ibp - c) \tag{3.9}$$

where:

- p represents path momentum,
- σ^2 is the diffusion coefficient,
- b is a drift parameter,

- c is a potential term.

The action of a path is defined as

$$S_{\text{path}} = \sum_{t=1}^{N} H(p_t) \tag{3.10}$$

The final output using the Feynman formulation is:

$$u(x_N) = \sum_{\text{paths}} e^{-S_{\text{path}}} u(x_0) \tag{3.11}$$

Paths with lower action contribute more significantly to the final representation.

Energy-Regularized Training Objective

An energy-based regularization term can be introduced into the loss:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{task}} + \lambda \sum_{\text{paths}} S_{\text{path}} \tag{3.12}$$

Alternatively, energy scaling can be integrated into each residual update:

$$u_{t+1} = u_t + \alpha_t F(u_t) \tag{3.13}$$

where

$$\alpha_t = e^{-H_t} \tag{3.14}$$

Continuous-Time Interpretation

As the network depth increases, the model approaches a continuous-time limit:

$$\frac{\partial u}{\partial t} = \sigma^2 \Delta u + b \nabla u - cu \tag{3.15}$$

This establishes a theoretical connection between deep residual networks and diffusion-based dynamical systems.

3.3 Summary

This chapter describes the methodology and design of leukemia cell classification using deep learning models. The proposal system requires a high-quality microscopic image, expert labeling, and preprocessing techniques such as noise removal, data augmentation, and normalization. Also, this system requires both functional and non functional requirements for higher accuracy. Several models are discussed, including SVM, ResNet, CNN, Decision Tree, Random Forest and KNN. SVMs are suitable for small and high-dimensional datasets. But ResNet and CNN can extract deep features from images. Decision trees and random forests give interpretable and robust classification, and KNN classifies data based on similarity.

Chapter 4

Implementation and Results

4.1 Environment Setup

The proposed system is developed using Python along with widely used deep learning frameworks. The implementation is carried out in a Jupyter Notebook environment, using libraries such as PyTorch, Torchvision, Scikit-learn, NumPy, and Matplotlib.

The system can run on either a GPU-enabled machine for faster computation or a standard CPU-based setup. It is compatible with both Windows and Linux operating systems. Pretrained models are utilized through Torchvision, while image preprocessing is handled using transformation pipelines to ensure consistency across the dataset.

4.2 Testing and Evaluation

The trained model is evaluated on a validation dataset consisting of previously unseen microscopic blood smear images. To assess performance, several standard evaluation metrics are used, including accuracy, precision, recall, F1-score, and the confusion matrix.

To address class imbalance, a weighted loss function is applied during training. The dataset is divided into 80% for training and 20% for validation. Throughout the training process, model performance is continuously monitored to reduce the risk of overfitting and ensure stable learning.

4.3 Results And Discussion

The results demonstrate that deep learning models, particularly ResNet50 are effective in classifying leukemia subtypes from microscopic images. The model successfully capture important cellular features such as shape, texture, and nucleus structure. The performance of the proposed model was evaluated during both initial training and fine-tuning phases. The training results show a steady improvement in accuracy and F1-score, along with a gradual reduction in loss.

During the initial training phase, the model achieved an accuracy of 74.36% in the first epoch, which improved up to 84.70% by the fifth epoch. Similarly, the F1-score increased from 0.7296 to 0.8247, indicating better class-wise performance.

```
Epoch 1: Loss=0.5773, Acc=0.7436, F1=0.7296
Epoch 2: Loss=0.4814, Acc=0.8286, F1=0.7803
Epoch 3: Loss=0.4754, Acc=0.8513, F1=0.8182
Epoch 4: Loss=0.4515, Acc=0.8116, F1=0.7807
Epoch 5: Loss=0.4509, Acc=0.8470, F1=0.8247
```

Figure 4.1: Initial Training Phase performance metrics.

After fine-tuning the model, performance improved further. The accuracy reached up to 88.81%, while the F1-score peaked at approximately 0.8701, demonstrating the effectiveness of transfer learning and fine-tuning in improving model performance.

```
[FT] Epoch 1: Loss=0.4257, Acc=0.8626, F1=0.8389
[FT] Epoch 2: Loss=0.4017, Acc=0.8881, F1=0.8701
[FT] Epoch 3: Loss=0.3660, Acc=0.8626, F1=0.8474
[FT] Epoch 4: Loss=0.3355, Acc=0.8725, F1=0.8542
[FT] Epoch 5: Loss=0.3033, Acc=0.8442, F1=0.8297
```

Figure 4.2: Fine-tuning Phase performance metrics.

4.4 Summary

This chapter outlined the implementation and evaluation of the proposed leukemia classification system. The model was trained using a structured pipeline and evaluated with standard performance metrics. The findings confirm that, with proper preprocessing and model optimization, deep learning approaches can effectively classify leukemia subtypes and support clinical decision-making.

Chapter 5

Standards and Design Constraints

5.1 Standards Compliance

In this project, we developed a deep learning based leukemia classification system using microscopic blood smear images. Since this system is related to medical diagnosis, maintaining proper standards is very important. We followed both academic research standards and software engineering standards during the development process. First, we followed a standard machine learning development pipeline. This includes data collection, preprocessing, model selection, training, validation, testing, and performance evaluation. This organized approach help to ensure the system is consistent and reliable. To evaluate the model performance, we used widely accepted metrics such as accuracy, precision, recall, F1 score and confusion matrix. These metrics are commonly used in medical image classification studies. We also followed ethical research standards. No personal patient information was included in the dataset. The images were used only for research and educational purposes. The system is designed as a decision support tool for doctors, not as a replacement for professional medical diagnosis. We kept a proper record of dataset preprocessing, model setting and the training steps so that the work can be verified and improved in the future.

5.1.1 Software Standards

The software follows standard coding and medical data practices:

- **Programming and Frameworks:** We have written our code in Python, following Python Enhancement Proposal (PEP 8) guidelines for readability and maintainability. We use standard deep learning frameworks (like TensorFlow/Keras or PyTorch) for reliable mathematical computation, utilized.

- **Medical Data Handling:** Though we use anonymized datasets, but data processing follows Health Insurance Portability and Accountability Act (HIPAA) principles to ensure no Personally Identifiable Information (PII) is ever exposed.

- **Image Processing:** Our system handles standard image formats (JPEG, PNG, TIFF) using computer vision libraries like OpenCV and Python Imaging Library (PIL) for preprocessing tasks like Gaussian blurring and CLAHE.

5.1.2 Hardware Standards

Deep learning models, particularly Convolutional Neural Networks (CNNs) like ResNet, require specific hardware configurations for efficient training:

- **Compute Standards:** For training purposes Graphics Processing Units (GPUs) compliant with NVIDIA's CUDA and cuDNN standards, accelerating tensor operations and speeds up.

- **Deployment Hardware:** Our model is optimized to run on standard commercial off the shelf (COTS) computers that are available in clinical pathology labs, minimizing the need for highly specialized deployment hardware.

5.1.3 Communication Standards

For integrated into a larger hospital information system or deployed as a web service:

- **Web Protocols:** All data transfers or any client-server interaction for uploading images and receiving predictions use HTTP/HTTPS protocols to ensure data transmission.

- **Interoperability:** Future integration with Electronic Health Records (EHR) will align with Fast Healthcare Interoperability Resources (FHIR) standards, which allows the AI to communicate smoothly with existing healthcare networks.

5.2 Design Constraints

During the development of this project, we faced several limitations in the real world that affected our system design.

5.2.1 Economic Constraint

One major limitation was financial. Deep learning models like ResNet require high computational power especially GPU based systems which are expensive. As undergraduate students, we did not have access to advanced hardware. So we used transfer learning with a pretrained ResNet instead of training a deep network from the start. This helped reduce both computational cost and training time. Medical data collection is also expensive. Preparing blood smear slides, capturing microscopic images and labeling them by expert pathologists require laboratory facilities and professional support.

5.2.2 Technical Constraint

The dataset we used was limited compared to real hospital level data. Limited data increases the risk of overfitting in deep learning models. Another technical challenge was class imbalance between leukemia subtypes. Some types of leukemia had fewer images than others, which can affect the performance of the model. To solve these problems, we applied data augmentation methods like rotation, zooming and flipping. We also used dropout and regularization to reduce overfitting.

5.2.3 Environmental Constraint

Training deep neural networks uses a large amount electrical energy and long training hours increase both energy and carbon footprint. To reduce environmental impact, we:

- Used pretrained models instead of training from scratch
- Kept the number of training epochs low
- Applied early stopping
- Optimized batch size

These steps helped to reduce unnecessary power consumption.

5.2.4 Ethical and Health Constraints

Since the system is used in the medical field, incorrect prediction may cause serious health risks. A wrong prediction might delay treatment or cause mental stress for patients. To solve these problems, we carefully validated the model using multiple evaluation metrics. We also stress that the system should support doctors, not replace them. Patient privacy and data confidentiality were strictly protected throughout the project.

5.2.5 Political Constraint

Healthcare data and automated diagnostic system are regulated by regional and national privacy laws. To navigate these political constraints and protect patient confidentiality, our system strictly uses anonymized datasets.

5.2.6 Sustainability and Social Impact

This project has important social value, especially in developing countries like Bangladesh where there are few hematologists experts. Manual blood smear analysis takes a lot of time and requires experienced specialists. In rural areas, access to such specialists is limited. Our automated classification system can help doctors by providing faster initial results. In terms of sustainability, the system can be scaled and improved in the future. For example:

- The dataset size can be made larger through hospital collaborations.
- More leukemia subtypes can be included.
- Advanced architectures and Attention mechanisms can be added.
- This system developed as a web or cloud based application.

This makes the project suitable for long term research and practical use.

5.3 Cost Analysis of AI-Based based Leukemia Subtype Classification System

Table: Cost Analysis of Leukemia Subtype Classification System

Cost Items	**Unit Price (Tk)**	**Units**	**Sub Total (Tk)**	**Assigned Members**
Dataset Collection & Annotation	60	10	600	Sajedur, Selim
Data Preprocessing & Augmentation	50	12	600	Tasnema, Tanvin
Model Development (CNN, ResNet, ML Models)	70	15	1050	Selim, Afia
Model Training & Hyperparameter Tuning	65	20	1300	Tasnema, Sajedur
Performance Evaluation & Testing	55	12	660	Tanvin, Tasnema
Backend Development (API for Model Deployment)	75	10	750	Selim, Afia
Frontend Interface Development	60	8	480	Tanvin, Sajedur

System Integration	60	8	480	Afia, Sajedur
Cloud Deployment and GPU Usage	120	4	480	Selim, Afia
Documentation (SRS, Report Writing)	40	15	600	Sajedur, Tanvin
Presentation & Miscellaneous	–	–	500	Afia, Selim
Total Estimated Cost			7500 Tk	

5.4 Complex Engineering Problem

Developing an automated leukemia classification system is a profound engineering challenge at the intersection of machine learning, medical imaging, and clinical ethics. It goes far beyond basic image recognition. The process demands meticulous preprocessing of microscopic blood smears, resolving imbalanced datasets, and fine-tuning deep learning models to meet rigorous medical standards. Most importantly, because this tool is built to empower hematologists rather than replace them, patient safety, system reliability, and clear interpretability must remain at the very core of its design

5.4.1 Knowledge Profile

Table: Mapping with Knowledge Profile (BAETE WK1–WK5)

Knowledge Profile	**Description**
WK1: Mathematics	Training a deep neural network is an exercise of mathematics. We relied on linear algebra to process high-dimensional image data through complex tensor operations, probability theory guided the final classification step by using a Softmax function to calculate exact likelihood of specific leukemia subtypes, and statistical evaluation metrics for model training and performance analysis.

WK2: Natural Sciences	A purely computational approach falls short in medical diagnostics without a solid foundation in hematology. To successfully teach a model to differentiate healthy white blood cells from four distinct leukemia subtypes (ALL, AML, CLL, and CML), we first needed to understand cellular morphology ourselves. Recognizing how a cell's pathology is dictated by its nucleus shape, chromatin density, and cytoplasm texture directly shaped our preprocessing strategy. Basic understanding of biological and medical science related to blood cell morphology and leukemia pathology.
WK3: Engineering Fundamentals	We engineered efficient data pipelines capable of managing massive batches of high-resolution microscopic images without overwhelming system memory. By applying core algorithmic thinking and data structure management, we also built and evaluated traditional machine learning baselines—such as Random Forests and Support Vector Machines—to establish performance benchmarks. Ultimately, these foundational programming practices kept our codebase clean, scalable, and fully optimized for cloud-based GPU execution..
WK4: Specialized Engineering Knowledge	Our project pushed into highly specialized territory, merging advanced computer vision with deep learning. We focused on Convolutional Neural Networks (CNNs), specifically a pre trained ResNet architecture to extract deep morphological features while avoiding the vanishing gradient problem. To prepare our complex medical data for this network, we applied targeted image processing techniques using OpenCV such as Gaussian blurring to cut through noise and CLAHE to amplify subtle cellular structures. Finally, we integrated Grad-CAM to bring visual interpretability (Explainable AI) to the model.

WK5: Engineering Design	We architected a complete, end-to-end automated classification framework. This meant designing a seamless pipeline that flowed from raw data ingestion and synthetic augmentation tackling severe class imbalances head on up to feature extraction, model training, and final evaluation. Because we built this system for real world clinical environments, we designed it around critical medical constraints. By prioritizing metrics like high Recall (sensitivity), we actively minimized the risk of dangerous false negative diagnoses, ensuring the final product acts as a safe, reliable decision-support tool for healthcare professionals.

5.4.2 Complex Problem Solving

Our proposed work is analyzed according to the Complex Engineering Problem (CEP) attributes defined by Board of Accreditation for Engineering & Technical Education (BAETE). This project includes multidisciplinary knowledge, advanced analytical techniques, ethical considerations, and stakeholder involvement. The following mapping shows how the project satisfies the BAETE defined attributes of complex engineering problems (P1–P4) with justification for each category.

Mapping with complex problem solving

Table: Mapping with Complex Problem Solving

Problem Attribute	**Description**
P1: Depth of Knowledge Required	The project requires multidisciplinary knowledge artificial intelligence, machine learning and deep learning, medical image processing, and fundamental understanding of hematology and leukemia subtypes. It also involves advanced mathematical concepts such as linear algebra, probability, optimization algorithms, and statistical evaluation metrics.
P2: Depth of Analysis Required	The project requires extensive preprocessing, feature extraction, model selection, hyperparameter tuning, and performance evaluation. Comparative analysis among multiple machine learning and deep learning models is necessary to determine optimal results.

P3: Familiarity with Issues	Automated leukemia subtype classification isn't a routine engineering task. The problem involves handling variability in medical images, data noise, class imbalance, and interpretability concerns, requiring innovative and adaptive problem-solving approaches.
P4: Extent of Stakeholder Involvement	Key stakeholders include medical professionals, pathologists, hospital authorities, researchers, and patients. Their involvement is essential to ensure clinical relevance, system usability, and reliability of diagnostic support.

5.4.3 Engineering Activities

Our proposed work is further assessed based on the Complex Engineering Activities framework defined by the BAETE. The development of an automated leukemia subtype classification system involves diverse technical resources, multidisciplinary interaction, innovation, societal implications, and non-routine problem-solving. The following table presents the mapping of the project with respect to the Complex Engineering Activity attributes (A1–A4) along with appropriate justification.

Attribute	**Description**
A1: Range of Resources	The project requires diverse resources microscopic blood smear image datasets from hospitals and publicly available sources, high performance computing resources (GPUs) for training deep learning models, machine learning frameworks (TensorFlow or PyTorch), and domain expertise from medical professionals for validation and interpretation of results.
A2: Level of Interaction	The project demands interaction between multidisciplinary domains computer science, artificial intelligence, and medical science. High quality labeled medical data is essential for achieving reliable classification accuracy. Additionally, significant computational processing power is required for model training and optimization.
A3: Innovation	The project applies machine learning and deep learning principles innovatively to automate leukemia subtype classification from microscopic images. The integration of advanced CNN-based architectures and comparative model analysis introduces a data-driven approach to assist traditional diagnostic methods.

A4: Familiarity	While medical image analysis and machine learning techniques exist individually, their integration for automated leukemia subtype classification presents a non-trivial and non routine complex engineering challenge. The task requires adaptation of advanced deep learning models to domain-specific medical imaging constraints.

5.5 Summary

This chapter outlined the standards and constraints guiding the leukemia classification model. Following strict software, hardware, and communication standards ensures the system is technically sound. By addressing economic, ethical, health, and sustainability factors, we designed a safe, reliable tool ready for real world medical use. Ultimately, tackling these complex engineering challenges ensures that the deep learning model is both academically rigorous and clinically valuable.

Chapter 6

Conclusion and Future Work

6.1 Summary

This project aimed to develop an automated, deep-learning system designed to classify leukemia cells into five distinct categories: healthy, Acute Lymphoblastic Leukemia (ALL), Acute Myeloid Leukemia (AML), Chronic Lymphocytic Leukemia (CLL), and Chronic Myeloid Leukemia (CML). Traditionally, diagnosing these conditions requires manual microscopic examination which is a slow process that relies heavily on the pathologists.

To modernize this workflow, we developed a structured diagnostic pipeline. We first applied image preprocessing techniques, such as Gaussian blurring and CLAHE, to bring cellular details into sharper focus. From there, we utilized data augmentation to overcome natural class imbalances and enrich the dataset.

For feature extraction and classification, we implemented the ResNet50, which demonstrated strong performance at capturing critical cellular structures. The model showed steady improvement during initial training, achieving an accuracy of 84.70% by the fifth epoch. Ultimately, these findings underscore the potential of AI based models to act as reliable decision support tools helping hematologists work more efficiently by delivering rapid initial screenings.

6.2 Limitations

While the initial results are highly encouraging, the current system does face a few practical constraints:

- **Data Size and Imbalance:** Our dataset was relatively small compared to what one might find in a real hospital environment, slightly elevating the risk of model overfitting. Additionally, natural class imbalances where some leukemia subtypes were underrepresented required us to rely on synthetic data generation to fill in the gaps.

- **Resource Constraints:** Deep learning architectures like ResNet are computationally hungry. They demand expensive, GPU-based systems, which made continuous model training and extensive hyperparameter tuning difficult for a resource-constrained undergraduate team.
- **Hardware Dependencies:** The model fundamentally relies on high-resolution microscopic blood smear images, meaning specialized laboratory equipment is still required to capture the initial data.
- **Clinical Risks:** Because this technology operates in the medical field, the stakes are high. A false classification could delay critical treatments or cause unnecessary distress for patients. For this reason, the system is designed strictly as a supplementary decision-support tool, not a replacement for a professional medical diagnosis.

6.3 Future Work

Looking ahead, our goal is to make the leukemia classification system more robust and ready for real-world application. Future development will focus on the following key areas:

- **Dataset Expansion:** We plan to collaborate with hospitals to build a larger, multi-center dataset. This will significantly improve the model's robustness and its ability to generalize across different clinical environments.
- **Architectural Enhancements:** We intend to integrate more advanced neural network architectures and Attention mechanisms. These upgrades will help the model extract highly complex features and further improve its multiclass accuracy.
- **System Deployment:** To maximize the project's social impact—especially in rural or underserved areas—we aim to deploy the system as a web- or cloud-based application. We will also explore integrating it with Electronic Health Records (EHR) using FHIR standards to ensure smooth communication with existing healthcare networks.
- **Explainable AI (XAI):** Building trust with healthcare professionals is paramount. To achieve this, we will prioritize integrating Gradient-weighted Class Activation Mapping (Grad-CAM). This will generate visual heatmaps highlighting exactly which regions of a cell influenced the model's predictions, making the AI's decision-making process transparent and clinically interpretable.